Card Fraud Triage via Gaussian Mixture Density Scoring: Phase 1 — Machine Learning Evaluation

Team T033, Section A, Project 26 — Rishihood University

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***Abstract*—We treat credit card fraud as an anomaly detection problem. Every model is trained only on legitimate transactions, and each new transaction is scored by how unusual it looks. We compare five methods on the ULB Credit Card Fraud dataset: a Gaussian Mixture Model (GMM), PCA reconstruction error, Isolation Forest, One-Class SVM and Local Outlier Factor. The test set has 56,962 transactions, of which 75 are fraud (0.13%). PCA reconstruction error is the best method, with a test PR-AUC of 0.7425. It catches 77.3% of test frauds when only 0.5% of legitimate transactions may be flagged. The GMM reaches a test PR-AUC of 0.4142, but its validation score (0.1950) is much lower, which shows that it is sensitive to its settings and to the small number of frauds. We report precision-recall curves, recall at fixed false-positive budgets, precision at review capacity, bootstrap confidence intervals and threshold analysis. A deep autoencoder will be added in Phase 2.**

***Index Terms*—anomaly detection, fraud detection, Gaussian mixture model, PCA reconstruction, class imbalance, precision-recall**

I. Introduction

Banks must decide which card transactions to check first. Three facts make this hard for ordinary supervised learning.

1) **Fraud is very rare.** In the European card dataset [1], only 492 of 284,807 transactions (0.1727%) are fraud. A model that always says "legitimate" is 99.83% accurate and finds no fraud at all.
2) **Labels arrive late.** A transaction is confirmed as fraud only days or weeks later, after the cardholder disputes it [2].
3) **Behaviour changes.** Fraudsters change their methods, and normal spending also changes over time [2].

Because of this, we learn what normal transactions look like and flag those that differ. Fraud labels are used only to choose settings and to measure results.

A. Scope and Contributions

Phase 1 covers classical machine learning. We study a GMM fitted with the Expectation-Maximization (EM) algorithm [3] and four baselines: PCA reconstruction error [4], Local Outlier Factor (LOF) [5], One-Class SVM [6] and Isolation Forest [7]. Deep autoencoders are left for Phase 2. Our main contributions are:

- A time-ordered split (60% train, 20% validation, 20% test) in which scalers and models see only legitimate training data.
- A check of how far the features are from a normal (Gaussian) shape.
- An evaluation built around operational limits: PR-AUC, recall at fixed false-positive rates, and precision at a fixed review capacity.
- A study of how the GMM choices (covariance type, number of components K, regularization) change fraud detection.

II. Related Work

A. Gaussian Mixture Models

A GMM describes the data as a weighted sum of K Gaussian components:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \qquad \sum_{k=1}^{K} \pi_k = 1. \tag{1}$$

The parameters are found with EM [3]. The anomaly score is the negative log-likelihood, $s(\mathbf{x}) = -\ln p(\mathbf{x})$, so unlikely points get high scores. The number of components is usually chosen with the Akaike criterion (AIC) [8] or the Bayesian criterion (BIC) [9].

B. Baseline Detectors

- **Isolation Forest** [7] cuts the data with random splits. Unusual points are separated after fewer cuts.
- **One-Class SVM** [6] draws a boundary around normal data in a kernel space.
- **PCA reconstruction error** [4] keeps the main principal components and measures how badly a point is rebuilt from them.
- **Local Outlier Factor** [5] compares the density around a point with the density around its neighbours.

Table I summarizes the methods.

TABLE I
ANOMALY DETECTION METHODS USED IN THIS WORK.

Method	Score	Known weakness
GMM	Negative log-likelihood	Needs near-Gaussian shapes
Isolation Forest	Short isolation path	Weak in many dimensions
One-Class SVM	Distance to boundary	Slow on large data
PCA	Reconstruction error	Linear structure only
LOF	Local density ratio	Slow queries

TABLE II
TIME-ORDERED SPLIT AND FRAUD RATE.

Split	Rows	Legit	Fraud	Rate
Train	170,884	170,524	360	0.2107%
Validation	56,961	56,904	57	0.1001%
Test	56,962	56,887	75	0.1317%
Total	284,807	284,315	492	0.1727%

C. Evaluation with Rare Fraud

ROC-AUC can look excellent even for a weak detector. With about 57,000 legitimate transactions in the test set, hundreds of false alarms still give a false-positive rate below 1% [10], [11]. Precision-recall curves react directly to false alarms, so we use PR-AUC as the main metric. We also report recall at fixed false-positive rates and precision when only a fixed number of alerts can be reviewed.

III. DATA AND EXPLORATORY ANALYSIS

A. Dataset and Split

The dataset has 284,807 transactions with 31 numeric columns, collected over 48 hours in September 2013. There are no missing values. We found 1,081 duplicate rows (0.38%) and kept them. The data are cut by time into train, validation and test sets (Table II). The fraud rate differs between the periods, so shuffled cross-validation would hide this change over time.

B. What the Data Look Like

- **Amount.** Legitimate amounts average \$88.29 (median \$22.00, maximum \$25,691). Fraud amounts average \$122.21 (median \$9.25, maximum \$2,126). Amounts are strongly skewed, so we use $\ln(1 + \text{Amount})$.
- **Separating features.** Two-sample Kolmogorov-Smirnov (KS) tests show that fraud differs most on V14 (KS = 0.854), V10 (0.832), V12 (0.800), V17 (0.787) and V11 (0.785).
- **Correlation.** V1–V28 are principal components, so they are almost uncorrelated ($|r| < 0.15$ on legitimate training data).

C. Is the Data Gaussian?

A GMM assumes each cluster is roughly Gaussian. We tested this on the 28 V-features of legitimate training data.

TABLE III
FEATURE SETS COMPARED ON VALIDATION DATA ($K = 3$, DIAGONAL GMM).

Feature set	Dim.	Val PR-AUC
raw	30	0.04315
base	31	0.04659
robust Amount	31	0.04359
Yeo-Johnson Amount	31	0.04509
whitened	31	0.04659
norms	34	0.05928
engineered	**31**	**0.06934**

Shapiro-Wilk tests reject normality for all 28 ($p < 0.05$). Many features have very heavy tails: the excess kurtosis is 115.0 for V2, 77.0 for V7 and 261.7 for V8, and only 9 of 28 features have $|\text{kurtosis}| < 3$. A single Gaussian would give too little probability to extreme values, so we use a mixture of several components.

D. No-Leakage Rules

All scalers and transforms are fitted on legitimate training rows only and then frozen. Fraud labels are never used to fit a model. Validation data are used to choose settings and thresholds, and the test set is used once at the end.

IV. FEATURE ENGINEERING

The V-features are anonymous and already uncorrelated, so we cannot build meaningful crosses between them. We therefore focused on four things: a better Amount scale, a cyclic encoding of time, summary norms of the V-features, and removing features with no signal.

We tested seven feature sets with a simple $K = 3$ diagonal GMM (Table III, Fig. 1):

- **raw** (30 features): V1–V28, raw Amount and Time.
- **base** (31): V1–V28, $\ln(1 + \text{Amount})$, and $\sin$, $\cos$ of the time of day.
- **robust / Yeo-Johnson**: base with a different Amount scaling.
- **whitened**: base with every V-feature rescaled to unit variance.
- **norms** (34): base plus the length $\|\mathbf{V}\|_2$, the number of $|V_i| > 3$, and $\max |V_i|$.
- **engineered** (31): the norm features, with V22, V23 and V25 removed because their KS value was below 0.05.

The log transform beat the robust and power transforms. Adding the norm features raised PR-AUC from 0.04659 to 0.05928 (+27.2%). Removing the three weak features raised it to 0.06934 (+48.8% over base). We use this 31-dimensional engineered set for the GMM. The baselines in Section VII use the base set (V1–V28, log amount, and sin and cos of time) and were not rerun on the engineered set.

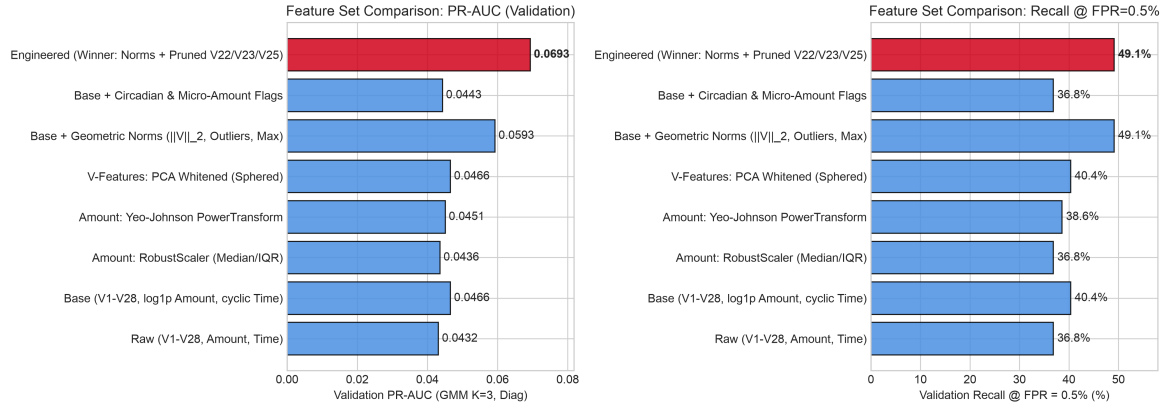


Fig. 1. Feature sets on the validation split. Left: PR-AUC. Right: recall at a 0.5% false-positive rate.

TABLE IV
NUMBER OF GMM PARAMETERS FOR $d = 31$ FEATURES.

Type	$K = 1$	$K = 5$	$K = 16$
Full	527	2,639	8,447
Tied	527	651	992
Diagonal	62	314	1,007
Spherical	32	164	527

TABLE V
TWO WAYS TO CHOOSE THE GMM.

Rule	**Cov.**	K / reg	**Val PR-AUC**
Lowest BIC	Full	16 / 10^{-6}	0.2123
Best validation PR-AUC	Full	12 / 10^{-2}	0.2684

V. GAUSSIAN MIXTURE MODEL THEORY

A. EM Algorithm

EM maximizes the likelihood of Eq. (1) by repeating two steps. The E-step computes, for each point $\mathbf{x}_n$, how much each component is responsible for it:

$$\gamma_{nk} = \frac{\pi_k \, \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \, \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}. \tag{2}$$

The M-step then updates the parameters, with $N_k = \sum_n \gamma_{nk}$:

$$\pi_k = \frac{N_k}{N}, \qquad \boldsymbol{\mu}_k = \frac{1}{N_k} \sum_n \gamma_{nk} \mathbf{x}_n, \tag{3}$$

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_n \gamma_{nk} (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^\top. \tag{4}$$

To stop a component from collapsing onto one point, a small value is added to the diagonal: $\boldsymbol{\Sigma}_k \leftarrow \boldsymbol{\Sigma}_k + \epsilon \mathbf{I}$. This value is the regularization (`reg_covar`).

B. Covariance Types

The covariance type controls how many parameters the model has (Table IV). Full covariance is flexible but needs many parameters. Diagonal covariance is much smaller and fits the uncorrelated V-features.

C. Choosing K: AIC and BIC

Both criteria reward a good fit and penalize extra parameters:

$$\text{AIC} = 2p - 2\ln \hat{L}, \qquad \text{BIC} = p \ln N - 2 \ln \hat{L}, \tag{5}$$

where p is the number of parameters, N the number of samples and $\hat{L}$ the maximized likelihood. For $N = 50{,}000$, $\ln N \approx 10.8$, so BIC penalizes extra components about five times more than AIC.

D. Limits of Density Scoring

1) A rare but honest transaction, such as a very large purchase, gets a high score and causes a false alarm.
2) Small fraud amounts look like normal shopping and get low scores.
3) Gaussian tails fall off quickly, so heavy-tailed legitimate spending is penalized.
4) In many dimensions, density values spread out and become less informative [12].

VI. GMM BASELINE RESULTS

A. Model Selection

We searched 120 settings: covariance type (full, tied, diagonal, spherical), K from 1 to 16, and regularization 10^{-6}, 10^{-4} or 10^{-2}. Figure 2 shows that AIC keeps decreasing as K grows, while BIC bends. Table V shows that the setting with the lowest BIC is not the best at finding fraud. The setting with the highest validation PR-AUC uses $K = 12$ with full covariance and regularization 10^{-2} (grid value 0.2684). We refit it on all 170,524 legitimate training rows. This final model has a validation PR-AUC of 0.1950.

B. Triage Performance

Table VI lists the GMM results. Thresholds come from validation data. On the test set the GMM reaches PR-AUC 0.4142, which is 314 times the base rate of 0.001317. At a 0.5% false-positive budget it raises 257 alerts and catches

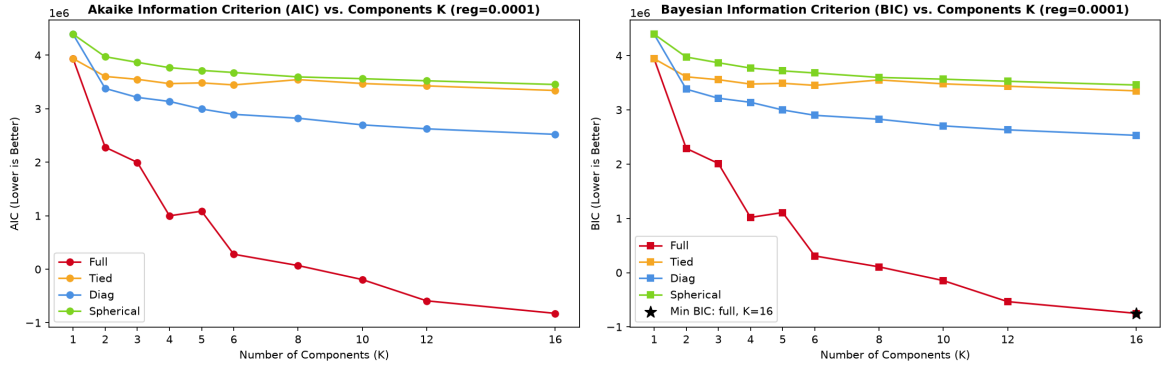


Fig. 2. AIC (left) and BIC (right) of the GMM against the number of components K, fitted on 50,000 legitimate training transactions.

TABLE VI
GMM TRIAGE PERFORMANCE.

Metric	Validation	Test
PR-AUC	0.1950	0.4142
ROC-AUC	0.9453	0.9274
Fraud rate	0.1001%	0.1317%
Recall at FPR 0.1%	24.56%	41.33%
Recall at FPR 0.5%	68.42%	69.33%
Recall at FPR 1.0%	75.44%	78.67%
Precision, 57 alerts (0.1%)	22.81%	47.37%
Precision, 285 alerts (0.5%)	13.33%	18.25%

52 of the 75 frauds (69.33%). When 285 transactions can be reviewed, 18.25% of them are fraud.

C. What Drives Scores and Misses

High scores for fraud come mostly from large values on V14, V10, V12 and the norm $\|\mathbf{V}\|_2$. False alarms come mostly from large amounts and from extreme values on V2. The lowest-scored frauds are small payments (about \$1.50 to \$22) that sit inside the normal region, so a density model cannot see them.

D. Stability Study

Figure 3 and Table VII show how the GMM reacts to its settings. These studies use a $K = 5$ model on the 31 engineered features, not the final $K = 12$ model.

- **Regularization.** With full covariance, the mean condition number of the covariances falls from about 10^7 at 10^{-6} to about 2×10^3 at 10^{-2}. Diagonal covariance falls from about 6×10^5 to about 2×10^2. Small regularization therefore gives badly conditioned matrices.
- **Random seed.** Over five seeds the validation PR-AUC is 0.0870 ± 0.0032 for the diagonal model, so the result is fairly stable.
- **Number of features.** PR-AUC rises from 0.0260 with 5 features to 0.0902 with 31 features, with a dip at 20 features (0.0582). The gain is not smooth.
- **Change over time.** The final model scores 0.1950 on validation but 0.4142 on test. With only 57 and 75 frauds, and with fraud rates that differ by period, single scores are noisy.

TABLE VII
VALIDATION PR-AUC AGAINST REGULARIZATION ($K = 5$).

Type	10^{-6}	10^{-4}	10^{-2}	10^{-1}
Diagonal	0.0790	0.0640	0.0702	0.0953
Full	0.7046	0.7034	0.6777	0.2838

The $K = 5$ full-covariance model in Table VII reaches a validation PR-AUC of about 0.70, much higher than the $K = 12$ model chosen by the grid (0.1950). We did not reconcile these runs in Phase 1. We list this as an open question for Phase 2.

VII. BASELINE DETECTORS

All baselines are trained on legitimate training rows only. Fraud labels are used only to pick settings on the validation split. They use the 31 base features.

Isolation Forest. The score is $s(\mathbf{x}, n) = 2^{-E[h(\mathbf{x})]/c(n)}$, where $h(\mathbf{x})$ is the path length needed to isolate $\mathbf{x}$ and $c(n)$ is the average path length for n points. We searched the number of trees (100, 200, 500), the sample size (256, 1024, auto) and the feature fraction (0.5, 1.0).

PCA reconstruction error. With the top k principal components $\mathbf{U}_k$ of legitimate data, the score is

$$s(\mathbf{x}) = \|\mathbf{x} - \mathbf{U}_k \mathbf{U}_k^\top \mathbf{x}\|^2. \tag{6}$$

We tried every k from 1 to 31 and kept the best on validation.

One-Class SVM. It uses an RBF kernel, and the score is the negative distance to the boundary. We searched $\nu \in \{0.001, 0.01, 0.05\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$. Because training is slow, we fitted it on a random sample of 40,000 legitimate rows.

Local Outlier Factor. We used novelty mode, fitted on 40,000 legitimate training rows, with $k \in \{10, 20, 50\}$.

Two reference scores help as sanity checks: a random score (its PR-AUC should equal the base rate) and the scaled transaction amount.

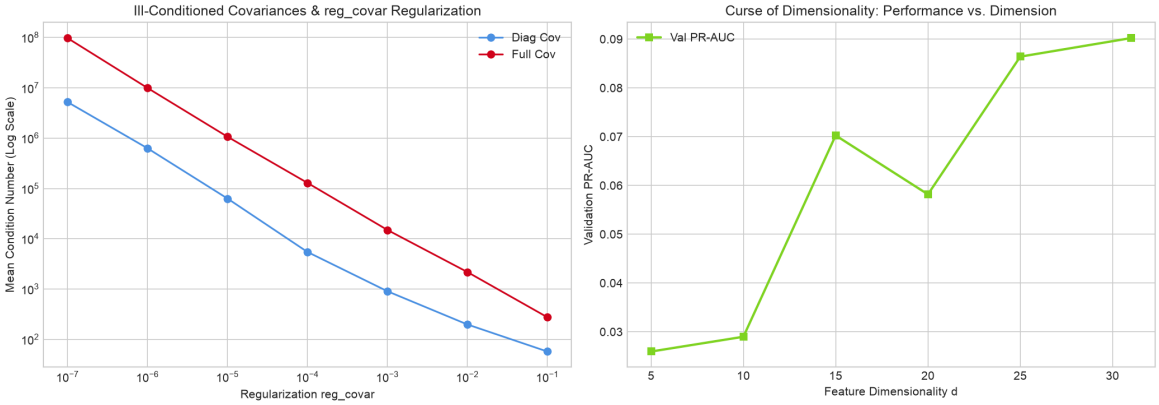


Fig. 3. GMM stability study. Left: condition number against regularization. Right: validation PR-AUC against the number of features.

TABLE VIII
TEST RESULTS. INTERVALS ARE 95% STRATIFIED BOOTSTRAP ($B = 2000$). THE TEST SET HAS 75 FRAUDS, SO THE INTERVALS ARE WIDE.

Model	PR-AUC [95% CI]	ROC-AUC	Recall @0.5%	Prec.@285
PCA	0.7425 [0.6415, 0.8368]	0.9303	0.7733	0.2000
LOF	0.4478 [0.3447, 0.5822]	0.9048	0.7200	0.1860
GMM	0.4142 [0.3050, 0.5403]	0.9274	0.6933	0.1825
OC-SVM	0.1322 [0.1030, 0.1784]	0.9604	0.6933	0.1649
IForest	0.0577 [0.0444, 0.0787]	0.9567	0.3467	0.0772
Amount	0.0034 [0.0026, 0.0051]	0.7095	0.0133	–
Random	0.0015 [0.0011, 0.0037]	0.4840	0.0133	–

VIII. Evaluation Protocol

- **PR-AUC** (average precision) is the main metric. It does not grow with the number of true negatives, so it reacts to false alarms.
- **Recall at a fixed false-positive rate (FPR).** The threshold is the $(1 - x)$ quantile of legitimate validation scores for $x \in \{0.1\%, 0.5\%, 1.0\%\}$. It is frozen and applied to the test set.
- **Precision at capacity.** The share of true frauds among the top K alerts.
- **Bootstrap intervals.** We draw 2,000 stratified resamples of the test set, resampling legitimate and fraud rows separately, and report the 2.5th and 97.5th percentiles.

IX. Results

A. Model Comparison

Table VIII ranks the models by test PR-AUC. PCA is clearly the best, and the other models are significantly worse than PCA under a paired bootstrap ($p < 0.05$). LOF and the GMM overlap, so we cannot tell them apart with 75 test frauds. Isolation Forest and One-Class SVM have ROC-AUC above 0.95, yet their PR-AUC is only 0.0577 and 0.1322. This shows why ROC-AUC alone is misleading here.

B. Precision-Recall Curves

Figure 4 shows the curves. A random ranker sits at the base rate of about 0.0013. The PCA curve stays high over a wide range of recall.

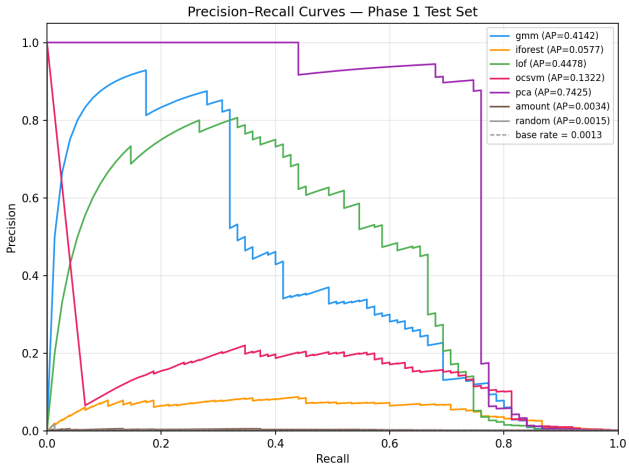


Fig. 4. Precision-recall curves on the test set. The dashed line is the base rate (0.132%).

TABLE IX
PCA AT FIXED FALSE-POSITIVE BUDGETS (TEST SET).

Budget	Alerts	TP	FP	Prec.	Recall
0.1%	141	57	84	0.404	0.760
0.5%	390	58	332	0.149	0.773
1.0%	750	58	692	0.077	0.773

C. Thresholds for the Best Model

Table IX shows PCA at three false-positive budgets. Thresholds come from validation and are frozen. Going from a 0.1% to a 0.5% budget adds 249 alerts but only one more fraud. Because the thresholds come from validation, the test false-positive rate can differ slightly from the budget.

D. Cost View

Figure 5 shows the expected cost per transaction when a missed fraud costs r times a false alarm. The cost ratio r is

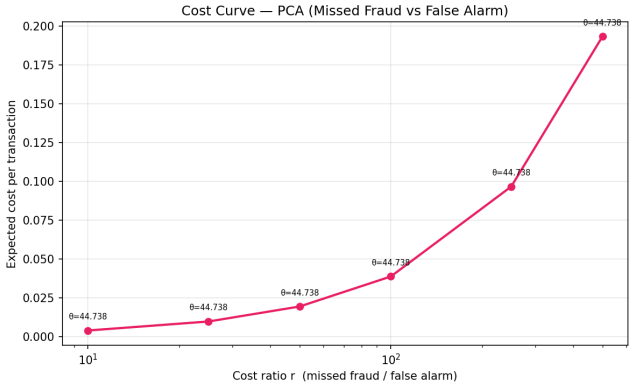


Fig. 5. Expected cost per transaction against the cost ratio r for PCA.

an assumption, so this figure shows how the best threshold depends on the business, not a recommended setting.

X. Discussion and Limitations

Why PCA wins. The V-features are themselves principal components. Legitimate transactions can be rebuilt well from a few components, while fraud follows a different pattern and is rebuilt badly.

Why the GMM is unstable. Its validation score (0.1950) and test score (0.4142) differ a lot. Only 57 validation frauds are available to choose settings, and the results depend on K, the regularization and the time period.

Why Isolation Forest is weak. With 31 features, random axis-aligned cuts do not separate fraud well. Its PR-AUC (0.0577) is close to a random ranker.

False-alert burden. At the 0.5% budget, PCA raises 390 alerts on the test set. Of these, 58 are fraud and 332 are false alarms. At the 0.1% budget it raises 141 alerts and still finds 57 of the 75 frauds.

Limits of this study.

- The GMM uses the engineered features and the baselines use the base features, so the feature input is not identical across models.
- Only 75 test frauds, so the intervals are wide and LOF and GMM cannot be separated.
- The features are anonymous, so we cannot add domain features.
- The data cover only 48 hours and one card network.
- Cost ratios are assumed and are specific to each bank.
- The $K = 5$ full-covariance result noted above is unresolved.

This is a research prototype, not a production fraud system.

XI. Conclusion and Future Work

We compared five unsupervised detectors for card fraud triage. PCA reconstruction error was best on the time-ordered test set (PR-AUC 0.7425). The GMM was in the middle (0.4142) and was sensitive to its settings and to the few available frauds. In Phase 2 we will train an undercomplete or denoising autoencoder on the same split and compare it with the GMM and PCA, using the same metrics.

Individual Contributions

- **Kushal Sarkar:** project setup, data pipeline, metrics module, integration, final comparison, report assembly.
- **Prachee Dhar:** literature review and EDA (duplicates, imbalance, KS drift, Gaussianity checks).
- **Apoorva Choudhary:** feature engineering, GMM grid search, AIC/BIC analysis, EM theory, stability study.
- **Vani Rudra:** baselines (PCA, Isolation Forest, OC-SVM, LOF), hyperparameter sweeps, theory sections.
- **Krishna Pramod Dubale:** evaluation toolkit, bootstrap intervals, threshold policies, cost curves, master table.

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